THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Alkanes are: 1

(a) alcohols **(b)** acids

(c) saturated hydrocarbons **(d)** unsaturated hydrocarbons

2. Burning of carbon compounds in oxygen is called: 1

(a) esterification **(b)** combustion **(c)** hydrogenation **(d)** saponification

3. The formula of ethane is: 1

(a) CH_4 **(b)** C_3H_8 **(c)** C_2H_4 **(d)** C_2H_6

4. Which is a saturated hydrocarbon? 1

(a) propyne **(b)** propene **(c)** ethene **(d)** propane

5. Products of complete combustion of hydrocarbon are: 1

(a) CO and water **(b)** alcohol

(c) CO_2 and water **(d)** carbon and hydrogen

6. The formula of ethene is: 1

(a) C_2H_2 **(b)** C_2H_4 **(c)** C_2H_6 **(d)** CH_4

7. The functional group of alcohols is: 1

(a) -COOH **(b)** -OH **(c)** -Cl **(d)** -CHO

8. Ethanol is commonly known as: 1

(a) vinegar **(b)** soap **(c)** alcohol **(d)** ester

9. The formula of ethyne is: 1

(a) CH_4 **(b)** C_2H_4 **(c)** C_2H_2 **(d)** C_2H_6

10. The functional group of carboxylic acids is: 1

(a) -OH **(b)** -CHO **(c)** -COOH **(d)** C=O

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Name the functional groups in 2

(i) CH_3OH

(ii) CH_3COOH.

12. What is esterification? 2

13. What are saturated hydrocarbons? Give an example. 2

14. Name two allotropes of carbon. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the names of the first two alkynes with formulas. 3

16. State three uses of ethanoic acid. 3

17. What is scum? Why is it formed? 3

OR

Name the functional groups: -OH, -CHO, -COOH. 3

18. Write combustion of methane with the equation. 3

OR

Write the structure of ethane, ethene and ethyne. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Write the five properties of carbon compounds. 5

OR

Describe diamond, graphite and fullerene (structure, one property, one use). 5

20. Write the first six members of the alkane homologous series with formula. 5

OR

What are addition and substitution reactions? Give one equation each. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND